Stochastic Finance and Risk Modelling

Chapter 1: From Probability to Stochastic Process

Based on slides by Ioane Muni-Toke
Laurence Carassus and Gaoyue Guo

Expanded Study Resource

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Introduction

This document serves as a comprehensive study guide for the course "Stochastic Finance and Risk Modelling". It expands upon the lecture slides relative to Chapter 1: "From probability to stochastic process".

The goal of this text is to provide a rigorous mathematical foundation required for modern financial modeling. We begin with the axiomatic definition of probability spaces and σ-algebras, move through random variables and their distributions, and finally establish key convergence results and the concept of conditional expectation, which is central to martingale theory and asset pricing.

Chapter 1

Topological Prerequisites

To fully leverage the power of measure theory in probability (especially constructing Borel σ-algebras and understanding convergence), we need a solid understanding of **Topology**.

A topology generalizes the notion of "closeness" and "limits" without necessarily needing a distance function (metric), although in finance we mostly work with metric spaces.

1.1 Metric Spaces

Definition 1.1 (Metric Space)**.** A **metric space** is a pair (M, d) where M is a set and $d : M \times M \to [0, \infty)$ is a distance function satisfying:

1. **Identity**: $d(x, y) = 0 \iff x = y$.
2. **Symmetry**: $d(x, y) = d(y, x)$.
3. **Triangle Inequality**: $d(x, z) \leq d(x, y) + d(y, z)$.

The most important building block in a metric space is the **Open Ball**.

Definition 1.2 (Open Ball)**.** The open ball of radius $r > 0$ centered at $x \in M$ is:

$$B(x, r) := \{y \in M : d(x, y) < r\}.$$

In $\mathbb{R}$ with the Euclidean metric $d(x, y) = |x - y|$, the open ball is simply the open interval $(x - r, x + r)$. In $\mathbb{R}^2$, it is the interior of a disk.

1.2 General Topology

While metric spaces are intuitive, the rigorous definition of "measurable events" relies on the more general concept of **Open Sets**.

Definition 1.3 (Topology)**.** A **topology** on a set Ω is a collection $\mathcal{T}$ of subsets of Ω (called **open sets**) such that:

1. $\emptyset \in \mathcal{T}$ and $\Omega \in \mathcal{T}$.
2. Any **union** (finite or infinite) of open sets is open.
3. Any **finite intersection** of open sets is open.

In a metric space, a set U is open if for every point $x \in U$, there exists some small ball entirely contained in U ($B(x, \epsilon) \subset U$).

1.2.1 Closed Sets

A set F is **closed** if its complement $F^c = \Omega \setminus F$ is open. **Intuition**: Closed sets "contain their boundaries".

Example 1.1. In $\mathbb{R}$:

- $(0, 1)$ is open.
- $[0, 1]$ is closed.
- $[0, 1)$ is neither open nor closed.

1.2.2 Closure, Interior, Boundary

For any set A:

- **Interior** A°: The largest open set contained in A.
- **Closure** $\bar{A}$: The smallest closed set containing A.
- **Boundary** $\partial A = \bar{A} \setminus A^\circ$.

Crucially, a set A is closed if and only if $A = \bar{A}$.

1.3 Continuity

In calculus, we define continuity using $\epsilon - \delta$. In topology, we have a much cleaner definition that connects directly to σ-algebras.

Definition 1.4 (Continuous Function)**.** A function $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is **continuous** if the inverse image of every open set is open.

$$\forall V \in \mathcal{T}_Y, \quad f^{-1}(V) \in \mathcal{T}_X.$$

This definition explains why random variables are defined as measurable functions: variables are to σ-algebras what continuous functions are to topologies.

1.4 Important Properties for Probability

1.4.1 Separability

A topological space is **separable** if it contains a countable dense subset.

- $\mathbb{R}$ is separable because $\mathbb{Q}$ (rationals) is countable and dense in $\mathbb{R}$.
- **Why it matters**: Separability prevents the space from being "too big". It ensures that the Borel σ-algebra is well-behaved and that the support of a measure is unique.

1.4.2 Completeness

A metric space is **complete** if every Cauchy sequence converges to a limit within the space.

- $\mathbb{R}$ is complete (Cauchy sequences converge to real numbers).
- $\mathbb{Q}$ is not complete (a sequence of rationals can converge to $\sqrt{2} \notin \mathbb{Q}$).

1.4.3 Polish Spaces

A **Polish Space** is a separable, completely metrizable topological space.

- Examples: $\mathbb{R}$, $\mathbb{R}^d$, $C([0,T],\mathbb{R})$ (continuous functions, via Wiener measure).
- These are the "nice" spaces where modern probability theory lives.

1.4.4 Compactness

A set K is **compact** if every open cover has a finite subcover. In $\mathbb{R}^d$ (Heine-Borel theorem), a set is compact if and only if it is **closed** and **bounded**. Compactness is crucial for proving the existence of limits (Prohorov's Theorem talks about "tightness", which is compactness for measures).

Chapter 2

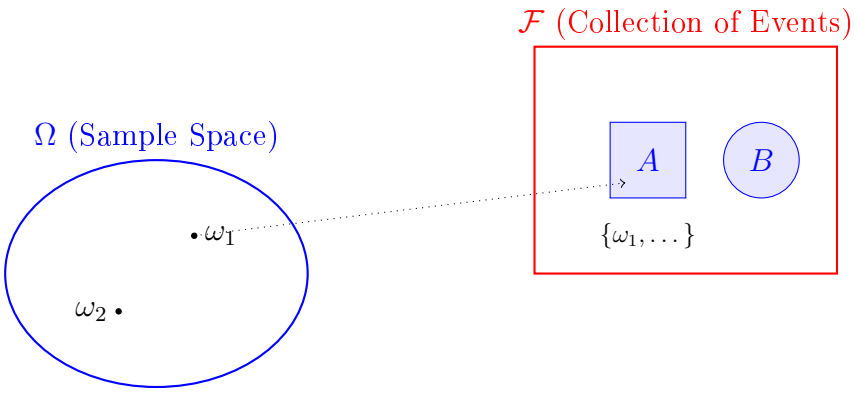
Probability Spaces and σ-algebras

2.1 Prerequisites: Set Theory & Notation

Before defining measurable spaces, let us clarify the distinction between a **set**, a **collection**, and the notation we use.

2.1.1 Set vs. Collection

- **Sample Space** Ω: The set of all possible outcomes ω.
- **Event** A: A subset of Ω ($A \subseteq \Omega$).
- **Collection** $\mathcal{F}$: A "set of sets". It contains events, not outcomes directly.



2.1.2 Example: Two Coin Tosses

Consider an experiment where we toss a coin twice.

- **Sample Space** $\Omega = \{HH, HT, TH, TT\}$ (4 outcomes).
- **Outcomes**: Simple elements like $\omega_1 = HH$.
- **Events**: Subsets of Ω, e.g., "Head on first toss" $A = \{HH, HT\}$.
- **Collection**: A σ-algebra $\mathcal{F}$ represents solvable questions.
  - If we know everything: $\mathcal{F} = 2^\Omega$ (contains $\{HH\}$, $\{HT\}$, etc.).

- If we only see the **first coin**: $\mathcal{F}_1 = \{\emptyset, \Omega, \{HH, HT\}, \{TH, TT\}\}$.
- Note that $\{HH\} \notin \mathcal{F}_1$ because we cannot distinguish HH from HT without seeing the second coin.

2.1.3 Countable vs. Uncountable

A set is **countable** (dénombrable) if you can list its elements in a sequence $x_1, x_2, x_3, \ldots$ (potentially infinite).

- **Countable**: Finite sets, $\mathbb{N}$ (integers), $\mathbb{Q}$ (rationals).
- **Uncountable**: $\mathbb{R}$ (reals), $[0, 1]$.

This distinction is crucial because σ-algebras are only closed under **countable** operations. We cannot sum probabilities over uncountable sets (e.g., $\sum_{x\in[0,1]} P(\{x\})$ makes no sense).

2.1.4 Key Notation

- **Difference** ($A \setminus B$ or $\Omega \setminus A$): "Everything in Ω except what is in A".

$$A^c := \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$$

- **Countable Union** ($\bigcup_{n\geq 1} A_n$): The union of a sequence $A_1, A_2, \ldots$. It means "at least one of the events A_n occurs".
- **Disjoint Union** ($\bigsqcup$ or "Pairwise Disjoint"): A union where no two sets overlap ($A_i \cap A_j = \emptyset$ for $i \neq j$). This is crucial for adding probabilities: $\mathbb{P}(\bigsqcup A_n) = \sum \mathbb{P}(A_n)$.
- **Power Set** (2^Ω): The set of *all* possible subsets of Ω. If $|\Omega| = n$, then $|2^\Omega| = 2^n$.

2.2 Measurable Spaces

The foundation of rigorous probability theory lies in measure theory. Before we can talk about probabilities of events, we must strictly define what an "event" is. Not every subset of a sample space can consistently be assigned a probability, especially when dealing with uncountably infinite sample spaces like the real line $\mathbb{R}$.

Definition 2.1 (σ-algebra)**.** Let Ω be a non-empty set (the sample space). A collection $\mathcal{F}$ of subsets of Ω (i.e., $\mathcal{F} \subseteq 2^\Omega$) is called a **$\sigma$-algebra** (or σ-field) on Ω if it satisfies the following three axioms:

1. **Empty set**: $\emptyset \in \mathcal{F}$.
2. **Complement**: If $A \in \mathcal{F}$, then its complement $A^c := \Omega \setminus A \in \mathcal{F}$.
3. **Countable Union**: If $(A_n)_{n\geq 1} \subset \mathcal{F}$ is a sequence of sets in $\mathcal{F}$, then their union is also in $\mathcal{F}$:

$$\bigcup_{n\geq 1} A_n \in \mathcal{F}.$$

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**. The elements of $\mathcal{F}$ are called **measurable sets** or, in the context of probability, **events**.

Remark 2.1. From these axioms, we can deduce other properties:

- The whole space Ω must be in $\mathcal{F}$ because $\Omega = \emptyset^c$.
- **Closed under Countable Operations**: The term "closed" means if you take any countable number of sets from $\mathcal{F}$ and apply set operations (union, intersection, complement), the result stays in $\mathcal{F}$.
- σ-algebras are closed under countable intersections. By De Morgan's laws:

$$\bigcap_{n\geq 1} A_n = \left(\bigcup_{n\geq 1} A_n^c\right)^c \in \mathcal{F}.$$

2.2.1 Examples of σ-algebras

1. **Trivial σ-algebra**: $\mathcal{F} = \{\emptyset, \Omega\}$. This is the smallest possible σ-algebra. It corresponds to having no information about the outcome of an experiment other than it happened.
2. **Power set**: $\mathcal{F} = 2^\Omega = \{A : A \subset \Omega\}$. This is the largest possible σ-algebra, containing every possible subset. While useful for discrete spaces (like rolling a die), it is often too large for continuous spaces like $\mathbb{R}$ to define a consistent measure (Vitali sets).
3. **Partition-generated**: Let $(A_i)_{i\in I}$ be a countable partition of Ω, meaning $\Omega = \bigsqcup_{i\in I} A_i$ (disjoint union). The σ-algebra generated by this partition consists of all possible unions of the sets A_i:

$$\mathcal{F} = \left\{\bigcup_{j\in J} A_j : J \subseteq I\right\}.$$

   This structure is very important in finance when modeling information flow in discrete time. If we know which partition element ω falls into, our information is described by this σ-algebra.

2.3 Generated σ-algebras

Often provided with a collection of sets of interest, we want to construct the smallest σ-algebra that allows us to measure them.

Definition 2.2 (Generated σ-algebra)**.** Let $\mathcal{C} \subseteq 2^\Omega$ be any collection of subsets of Ω. The σ-algebra generated by $\mathcal{C}$, denoted $\sigma(\mathcal{C})$, is the intersection of all σ-algebras containing $\mathcal{C}$:

$$\sigma(\mathcal{C}) := \bigcap_{\mathcal{F}\supset\mathcal{C}, \mathcal{F} \text{ is a } \sigma\text{-algebra}} \mathcal{F}.$$

It is the **smallest** σ-algebra containing $\mathcal{C}$.

2.3.1 Borel σ-algebra

In topological spaces, we want our algebraic structure (events) to be compatible with the topological structure (open sets, convergence).

Definition 2.3 (Borel σ-algebra)**.** Let (Ω, τ) be a topological space. The **Borel σ-algebra**, denoted $\mathcal{B}(\Omega)$, is the σ-algebra generated by the family of open sets $\mathcal{O}$ in Ω.

Case of $\mathbb{R}$ and $\mathbb{R}^d$

- On $\mathbb{R}$, $\mathcal{B}(\mathbb{R})$ is generated by open intervals $]a, b[$. It is the smallest family of sets containing all intervals that is closed under countable operations.
- On $\mathbb{R}^d$, $\mathcal{B}(\mathbb{R}^d)$ is generated by open rectangles (or "hyper-rectangles"):

$$\prod_{i=1}^{d}]a_i, b_i[= \{\omega = (\omega_1, \ldots, \omega_d) \in \mathbb{R}^d : a_i < \omega_i < b_i, \forall i\}.$$

Note that while $\mathcal{B}(\mathbb{R}^d)$ contains almost any set we can construct explicitly, there do exist **non-Borel sets**.

- **Non-Borel Sets**: These are "pathological" sets (like the Vitali set) constructed using the Axiom of Choice. They essentially "break" the concept of length/measure.
- We exclude them from our σ-algebra to ensure mathematical consistency.

Important Note: $\mathcal{B}(\mathbb{R}^d) \neq 2^{\mathbb{R}^d}$

- The Borel σ-algebra is strictly smaller than the Power Set.
- $\mathcal{B}(\mathbb{R})$ contains only sets constructed from open intervals.
- $2^{\mathbb{R}}$ contains **every** possible subset, including non-measurable ones.
- For probability theory, we work with $\mathcal{B}(\mathbb{R})$ to avoid paradoxes.

2.3.2 Product σ-algebra

When dealing with multiple random variables or stochastic processes, we work with Cartesian products of spaces.

Let $((\Omega_n, \mathcal{F}_n))_{n \geq 1}$ be a sequence of measurable spaces.

- **Finite Product**: For a finite number of spaces $\Omega_1 \times \cdots \times \Omega_n$, the product σ-algebra $\mathcal{F}_1 \otimes \cdots \otimes \mathcal{F}_n$ is generated by "rectangles" of the form:

$$A_1 \times \cdots \times A_n, \quad \text{where } A_i \in \mathcal{F}_i.$$

  Crucial Distinction:

  - $\Omega_1 \times \Omega_2$: Cartesian product of **points** (the ground space).
  - $\mathcal{F}_1 \otimes \mathcal{F}_2$: Sigma-algebra generated by rectangles. It is strictly larger than just $\mathcal{F}_1 \times \mathcal{F}_2$ (which is just a set of pairs of sets). The $\otimes$ symbol implies we generate all necessary unions/intersections to make it a valid σ-algebra.

Important property: $\mathcal{B}(\mathbb{R}) \otimes \cdots \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^n)$.

Example: Why do we need $\otimes$? (The Circle Problem) Consider the plane $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$.

- A "rectangle" is a set $A \times B$ (e.g., $x \in [0,1], y \in [0,1]$).
- A Circle $C = \{(x,y) : x^2 + y^2 \leq 1\}$ is a valid event we want to measure.
- **Problem**: A circle **cannot** be written as a single rectangle $A \times B$ or a finite union of them.
- **Solution**: The σ-algebra $\mathcal{F} \otimes \mathcal{F}$ allows for **countable unions**. We can approximate the circle with infinitely many small rectangles. Thus, the circle belongs to $\mathcal{F} \otimes \mathcal{F}$, even though it is not a "simple product" of sets.

- **Infinite Product**: For the infinite product space $\prod_{n\geq 1} \Omega_n$, the product σ-algebra $\bigotimes_{n\geq 1} \mathcal{F}_n$ is generated by **cylinders** with a finite basis. A measurable cylinder is a set of the form:
$$C = \{\omega \in \prod \Omega_n : \omega_{i_1} \in A_{i_1}, \ldots, \omega_{i_k} \in A_{i_k}\}$$
where only a finite number of coordinates are restricted. This is essential because we cannot define a uniform measure on infinite dimensions directly; we must work with these finite-dimensional projections (Kolmogorov Extension Theorem).

2.4 Probability Measures

Definition 2.4 (Measure)**.** A map $\mu : \mathcal{F} \to [0, \infty]$ is called a **measure** if:

1. $\mu(\emptyset) = 0$.
2. **σ-additivity**: For any sequence $(A_n)_{n\geq 1} \subset \mathcal{F}$ of **pairwise disjoint** sets,
$$\mu\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mu(A_n).$$

Definition 2.5 (Probability Measure)**.** A measure $\mathbb{P}$ is a **probability measure** if it satisfies the normalization axiom:
$$\mathbb{P}(\Omega) = 1.$$
The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**.

We denote the set of all probability measures on $(\Omega, \mathcal{F})$ as $\mathcal{P}(\Omega, \mathcal{F})$.

2.5 Support of a Measure

The concept of support formalizes the idea of "where the probability mass lives".

Definition 2.6 (Support)**.** The **support** of a measure μ, denoted $\text{supp}(\mu)$, is the closed set in Ω such that:

1. The measure of its complement is zero: $\mu(\Omega \setminus \text{supp}(\mu)) = 0$.

2. For any open set G such that $G \cap \operatorname{supp}(\mu) \neq \emptyset$, we have $\mu(G \cap \operatorname{supp}(\mu)) > 0$.

Essentially, it is the smallest closed set that carries all the mass of μ. If measurement values fall outside the support, it is an "impossible" event (probability 0).

For a Polish space (separable, completely metrizable, like $\mathbb{R}^d$), the support is unique and given by:

$$\operatorname{supp}(\mu) = \Omega \setminus \bigcup \{V : V \text{ is open and } \mu(V) = 0\}.$$

Chapter 3

Random Variables and Distributions

3.1 Random Variables

A random variable is, somewhat counter-intuitively, not a variable but a function. It connects the abstract world of the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to a concrete numerical space (usually $\mathbb{R}$ or $\mathbb{R}^d$) where we can perform calculations.

Definition 3.1 (Random Variable)**.** Let $(\Omega, \mathcal{F})$ be a measurable space and $(\Sigma, \mathcal{G})$ be another measurable space (often $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$). A map $X : \Omega \to \Sigma$ is called a **measurable function** or **random variable (r.v.)** if the pre-image of every measurable set in Σ is a measurable event in Ω:

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}, \quad \forall B \in \mathcal{G}.$$

Remark 3.1 (Notation)**.** In Definition 3.1, $(\Sigma, \mathcal{G})$ is the **target space** (or state space) where the values of the random variable live.

- Most of the time, $\Sigma = \mathbb{R}$ or $\mathbb{R}^d$.
- $\mathcal{G}$ is the corresponding Borel σ-algebra $\mathcal{B}(\mathbb{R})$ or $\mathcal{B}(\mathbb{R}^d)$.

So, the condition simply means: "The inverse image of any Borel set must be an event in $\mathcal{F}$".

3.1.1 Example: The Coin Toss Game

Consider a game where we toss a coin three times.

- **Sample Space** $\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.
- **Random Variable** X: Let X be the number of Heads.

$$X(HHH) = 3, \quad X(HHT) = 2, \quad \ldots, \quad X(TTT) = 0.$$

- **Measurability Check**: Is the question "What is the probability that $X = 2$?" well-defined?

- We check the pre-image:

$$X^{-1}(\{2\}) = \{\omega \in \Omega : X(\omega) = 2\} = \{HHT, HTH, THH\}.$$

  Since this set $\{HHT, HTH, THH\}$ is an event in our σ-algebra $\mathcal{F}$ (which is usually 2^Ω for finite spaces), then yes, X is a valid Random Variable.

This condition ensures that questions like "What is the probability that X falls in region B?" are well-defined, because $\{X \in B\}$ is an event in $\mathcal{F}$, on which $\mathbb{P}$ is defined.

3.2 Distributions and Pushforward Measures

When we observe a random variable X, we effectively transport the probability mass from Ω to the target space Σ.

Definition 3.2 (Pushforward Measure / Law)**.** Let X be a random variable from $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\Sigma, \mathcal{G})$. The **pushforward measure** of $\mathbb{P}$ by X, denoted $X_\#\mathbb{P}$ or $\mathbb{P}_X$, is a probability measure on $(\Sigma, \mathcal{G})$ defined by:

$$\mathbb{P}_X(B) := \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B), \quad \forall B \in \mathcal{G}.$$

This measure $\mathbb{P}_X$ is called the **law** or **distribution** of X.

In simpler terms, the distribution $\mathbb{P}_X$ tells us how the randomness is distributed on the value space (e.g., the real line), ignoring the underlying details of Ω.

3.3 Expectation and Integration

Definition 3.3 (Expectation)**.** The expectation of a random variable $X : \Omega \to \mathbb{R}^d$ is defined as the Lebesgue integral of X with respect to the measure $\mathbb{P}$:

$$\mathbb{E}[X] := \int_\Omega X(\omega)\, \mathbb{P}(d\omega).$$

3.3.1 Theorem of Transport (Transfer Theorem)

This fundamental theorem allows us to compute expectations without working directly with the abstract space Ω. We can work solely with the distribution of X on $\mathbb{R}^d$.

Theorem 3.2. *Let $X : \Omega \to \Sigma$ be a random variable and $g : \Sigma \to \mathbb{R}$ be a measurable function. Let $Y = g(X)$. Then Y is integrable (i.e., $Y \in L^1(\Omega)$) if and only if $g \in L^1(\Sigma, \mathbb{P}_X)$. In this case:*

$$\mathbb{E}[g(X)] = \int_\Omega g(X(\omega))\, \mathbb{P}(d\omega) = \int_\Sigma g(x)\, \mathbb{P}_X(dx).$$

This is the workhorse of probability. If X has a probability density function $f_X(x)$ on $\mathbb{R}$, this simplifies to the familiar Riemann integral:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx.$$

3.4 Moments

The "shape" of a random variable's distribution is often described by its moments.

Definition 3.4 (Moments)**.** Let X be a random variable.

- **Mean (1st Moment)**: $\mu = \mathbb{E}[X]$. Measures central tendency.
- **Variance (2nd Central Moment)**: $\sigma^2 = \mathbb{E}[(X-\mu)^2]$. Measures dispersion/risk.
- **Skewness (3rd Standardized Moment)**: $\gamma = \mathbb{E}[(\frac{X-\mu}{\sigma})^3]$. Measures asymmetry.
- **Kurtosis (4th Standardized Moment)**: $\kappa = \mathbb{E}[(\frac{X-\mu}{\sigma})^4]$. Measures "fat tails" (extreme events). Gaussian has $\kappa = 3$.

3.5 L^p Spaces

Definition 3.5 (L^p Spaces)**.** For $p \geq 1$, the space $L^p(\Omega, \mathcal{F}, \mathbb{P})$ consists of all random variables X such that $|X|^p$ is integrable:

$$\mathbb{E}[|X|^p] < \infty.$$

The norm is defined as $\|X\|_p = (\mathbb{E}[|X|^p])^{1/p}$.

- L^1: The space of random variables with **finite mean**.

  $$X \in L^1 \implies \mathbb{E}[|X|] < \infty.$$

  This is the minimal requirement to talk about "expected value".

- L^2: The space of random variables with **finite variance** (or finite energy/power).

  $$X \in L^2 \implies \mathbb{E}[X^2] < \infty.$$

  This is the most important space in finance because variance measures risk.

- **Hierarchy**: Since we are on a probability space (which has finite measure $\mathbb{P}(\Omega) = 1$), higher moments imply lower moments:

  $$L^p \subset L^q \quad \text{if } p > q.$$

  Specifically, $L^2 \subset L^1$. If a variable has finite variance, it definitely has a finite mean. The reverse is not true (e.g., Cauchy distribution).

3.5.1 Why is L^2 special? (Hilbert Space)

L^2 is the only L^p space that is a **Hilbert Space**. It comes equipped with an inner product:

$$\langle X, Y \rangle = \mathbb{E}[XY].$$

This allows us to talk about geometry:

- **Orthogonality**: Two random variables are orthogonal if $\mathbb{E}[XY] = 0$. If they are centered, this means they are **uncorrelated**.
- **Projections**: We can project a random variable onto a subspace. This is exactly what **Conditional Expectation** $\mathbb{E}[X|\mathcal{G}]$ is: the orthogonal projection of X onto the space of $\mathcal{G}$-measurable functions. This geometric interpretation is key to understanding martingales and optimal hedging.

3.5.2 Distinction: L^p Spaces vs σ-algebras $\mathcal{F}$

It is common to confuse these two concepts.

- **$\mathcal{F}$ is a collection of SETS (Events)**: It defines *what information is available.* "Can we answer the question 'Did event A happen?'".
- **L^p is a space of FUNCTIONS (Random Variables)**: It defines *random variables with certain properties* (finite energy/mean) that are **measurable** with respect to $\mathcal{F}$.
- **Relationship**: A function X belongs to $L^p(\Omega, \mathcal{F}, \mathbb{P})$ only if it is $\mathcal{F}$-measurable.

$$\mathcal{F} \text{ (Information)} \quad \leftrightarrow \quad L^p(\mathcal{F}) \text{ (Variables built on that information).}$$

Definition 3.6 (Variance and Covariance)**.** For $X, Y \in L^2(\Omega)$, the **covariance** is:

$$\mathrm{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The **variance** is $\mathrm{Var}(X) := \mathrm{Cov}(X, X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

3.6 Common Distributions

3.6.1 Discrete Distributions

1. **Binomial Distribution** $B(n, p)$: Describes the number of successes in n independent Bernoulli trials with probability p. Support: $\{0, 1, \ldots, n\}$.

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

2. **Poisson Distribution** $\mathcal{P}(\lambda)$: Describes rare events occurring at a known constant rate λ. Support: $\mathbb{N} = \{0, 1, 2, \ldots\}$.

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

3.6.2 Continuous Distributions

1. **Uniform Distribution** $\mathcal{U}(a, b)$: Constant probability density over an interval. Support: $[a, b]$.

$$\mathbb{P}_X(dx) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)\, dx.$$

2. **Gaussian (Normal) Distribution** $\mathcal{N}(m, \sigma^2)$: The most important distribution due to the Central Limit Theorem. Support: $\mathbb{R}$. parameters: mean m, variance σ^2.

$$\mathbb{P}_X(dx) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right) dx.$$

3.7 Gaussian Vectors

A random vector $X = (X_1, \ldots, X_d)$ is a **Gaussian vector** if any linear combination of its components is a univariate Gaussian method.

$$\forall a \in \mathbb{R}^d, \quad a^T X = \sum_{i=1}^{d} a_i X_i \sim \mathcal{N}(\mu_a, \sigma_a^2).$$

This is a much stronger condition than simply saying each marginal X_i is Gaussian. The law is determined by the mean vector $\mu = \mathbb{E}[X]$ and the covariance matrix Σ, where $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$.

Chapter 4

Independence and Convergence

4.1 Independence

Independence captures the intuitive idea that the outcome of one event does not influence the likelihood of another.

Definition 4.1 (Independence of Events)**.** Events $(A_\lambda)_{\lambda\in\Lambda}$ are **mutually independent** if for every finite sub-collection with indices $\lambda_1, \ldots, \lambda_k$, the probability of the intersection is the product of the probabilities:

$$\mathbb{P}\left(\bigcap_{j=1}^{k} A_{\lambda_j}\right) = \prod_{j=1}^{k} \mathbb{P}(A_{\lambda_j}).$$

Warning: Pairwise independence does NOT imply mutual independence. (Example: Bernstein's counterexample).

Definition 4.2 (Independence of σ-algebras)**.** Sub-σ-algebras $\mathcal{G}_1, \ldots, \mathcal{G}_n \subset \mathcal{F}$ are independent if for any choice of events $A_i \in \mathcal{G}_i$:

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} \mathbb{P}(A_i).$$

Definition 4.3 (Independence of Random Variables)**.** Random variables (X_λ) are independent if their generated σ-algebras $\sigma(X_\lambda) = X_\lambda^{-1}(\mathcal{G})$ are independent.

4.2 Modes of Convergence

In analysis, a sequence of numbers converges to a limit. In probability, since the random outcomes depend on ω, there are multiple ways to define "closeness" to a limit random variable Y.

Let $(Y_n)_{n\geq 1}$ and Y be random variables.

4.2.1 Almost Sure Convergence (a.s.)

Pointwise convergence for "almost all" outcomes.

$$Y_n \xrightarrow{\text{a.s.}} Y \iff \mathbb{P}\left(\{\omega : \lim_{n\to\infty} Y_n(\omega) = Y(\omega)\}\right) = 1.$$

This is the strongest form of pointwise convergence.

Decoding the Notation: The equation $\mathbb{P}(\{\omega : \lim Y_n(\omega) = Y(\omega)\}) = 1$ does **NOT** mean that any specific outcome ω has probability 1.

- We look at the **set of all "good" outcomes** $G = \{\omega \in \Omega : Y_n(\omega) \to Y(\omega)\}$.
- Construct this set first. For some ω, the sequence converges. For others, it might not.
- The condition says: The **probability of this set G is 1**.
- Equivalently, the set of "bad" outcomes $N = G^c$ has probability $\mathbb{P}(N) = 0$.
- *Example*: If we pick a number uniformly in $[0, 1]$, the probability of picking exactly 0.5 is 0. But the probability of picking *any* number in $[0, 1]$ is 1 itself.

4.2.2 L^p Convergence

Convergence in the mean of the p-th power difference.

$$Y_n \xrightarrow{L^p} Y \iff \|Y_n - Y\|_p = (\mathbb{E}[|Y_n - Y|^p])^{1/p} \to 0.$$

Implies L^q convergence for any $q < p$ (by Hölder's inequality).

Why is this useful?

- **Moment Convergence**: It implies that the moments converge: $\mathbb{E}[Y_n] \to \mathbb{E}[Y]$ and $\mathrm{Var}(Y_n) \to \mathrm{Var}(Y)$. (Note: a.s. convergence does NOT guarantee this, e.g., "mass escaping to infinity").
- **Stability**: It controls the "energy" of the error. In finance, L^2 convergence is crucial because the Itô Integral is constructed as a limit in L^2 space.
- **Completeness**: The space L^p is complete (Banach space). This means that if a sequence of random variables gets closer to each other (Cauchy), it is guaranteed to converge to a valid random variable within the space.

4.2.3 Convergence in Probability

The probability of observing a significant difference goes to zero.

$$Y_n \xrightarrow{\mathbb{P}} Y \iff \forall \epsilon > 0, \quad \mathbb{P}(|Y_n - Y| > \epsilon) \to 0.$$

Relations:

- $L^p \implies$ Probability (Markov inequality).
- Almost Sure $\implies$ Probability.
- The converses are generally false (though Probability $\implies$ a.s. along a subsequence).

4.2.4 Convergence in Law (Distribution)

Only the distributions approach the limit distribution. The random variables themselves might not be close.

$$Y_n \xrightarrow{\text{law}} Y \iff \mathbb{E}[f(Y_n)] \to \mathbb{E}[f(Y)] \quad \text{for all bounded continuous } f.$$

Also denoted as $\implies$ or $\xrightarrow{\mathcal{D}}$. This is the weakest form of convergence.

4.3 Limit Theorems

Let (X_n) be a sequence of independent and identically distributed (i.i.d.) random variables.

Theorem 4.1 (Strong Law of Large Numbers). *If $X_1 \in L^1$ (measurable mean $m = \mathbb{E}[X_1]$), then the empirical average converges almost surely to the theoretical mean:*

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} m.$$

Theorem 4.2 (Central Limit Theorem). *If $X_1 \in L^2$ (finite variance $\sigma^2 = Var(X_1)$), then the standardized sum converges in law to a standard Gaussian:*

$$\frac{\sqrt{n}(\bar{X}_n - m)}{\sigma} = \frac{\sum_{i=1}^{n}(X_i - m)}{\sigma\sqrt{n}} \xrightarrow{law} \mathcal{N}(0,1).$$

This explains the ubiquity of the Normal distribution in finance (returns aggregation).

Chapter 5

Conditional Expectation

Conditional expectation is arguably the most important concept in mathematical finance, as prices of fair assets are modeled as conditional expectations of their future payoffs under a specific measure (martingales).

5.1 Conditional Expectation in L^1

The elementary definition $\mathbb{E}[X|Y=y] = \sum x\mathbb{P}(X=x|Y=y)$ works for discrete variables but fails for continuous ones because $\mathbb{P}(Y=y)=0$. The rigorous definition relies on σ-algebras and measurability.

Theorem 5.1 (Existence and Uniqueness)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra (representing "partial information").*

*There exists a **unique** random variable Z, denoted by $\mathbb{E}[X|\mathcal{G}]$, such that:*

1. ***Measurability**: Z is $\mathcal{G}$-measurable.*
2. ***Partial Averaging Property**: For every bounded $\mathcal{G}$-measurable random variable H (or equivalently for all $A \in \mathcal{G}$):*

$$\mathbb{E}[ZH] = \mathbb{E}[XH] \quad \left(i.e., \int_A Z\, d\mathbb{P} = \int_A X\, d\mathbb{P} \right).$$

*This Z is called the **conditional expectation** of X given $\mathcal{G}$.*

Uniqueness holds $\mathbb{P}$-almost surely.

Intuition: What does $\mathcal{G}$-measurable mean? It means that the value of Z is fully determined by the information in $\mathcal{G}$.

- If $\mathcal{G}$ represents our knowledge at time t, then Z is a value we can **observe right now**.
- If $\mathcal{G} = \sigma(Y)$, then Z can be written as a deterministic function of Y: $Z = f(Y)$.
- In simple terms: "Given $\mathcal{G}$, there is no more randomness in Z".

Intuition: The "Partial Averaging" Concept Imagine $\mathcal{G}$ is generated by a partition of Ω into disjoint sets $A_1, A_2, \ldots$.

- On each set A_k, we cannot distinguish the specific outcomes ω. The available information is just "we are in A_k".
- Therefore, the best guess $Z = \mathbb{E}[X|\mathcal{G}]$ must be **constant** on A_k.
- To satisfy the property $\int_{A_k} Z d\mathbb{P} = \int_{A_k} X d\mathbb{P}$, this constant must be the **average value** of X on A_k:

$$Z(\omega) = \frac{1}{\mathbb{P}(A_k)} \int_{A_k} X(\omega') d\mathbb{P}(\omega') \quad \text{for } \omega \in A_k.$$

- In general σ-algebras (not just partitions), Z is a "smoothed" version of X that preserves the local averages over all $\mathcal{G}$-measurable sets.

Example: Die Roll Partition Consider rolling a fair die $\Omega = \{1, 2, 3, 4, 5, 6\}$.

- Let the information be "Is the number Even or Odd?". This generates a partition of Ω:

$$A_1 = \{1, 3, 5\} \text{ (Odd)}, \quad A_2 = \{2, 4, 6\} \text{ (Even)}.$$

- These sets are **disjoint** ($A_1 \cap A_2 = \emptyset$) and their union is the whole space Ω.
- If we know we are in A_2 (Even), our best guess for the outcome is the average of $\{2, 4, 6\}$, which is 4.
- So $\mathbb{E}[X|\text{"Even"}] = 4$.

Remark 5.2 (Generated σ-algebra)**.** If $\mathcal{G} = \sigma(Y)$, this is the **σ-algebra generated by** Y.

- It represents the information provided by observing the random variable Y.
- Formally, $\sigma(Y) = \{Y^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$.
- Conditioning on Y means conditioning on the information "we know the value of Y".

We write $\mathbb{E}[X|Y]$ instead of $\mathbb{E}[X|\sigma(Y)]$ for brevity.

5.2 Key Properties

Let $X, Y \in L^1$ and $\mathcal{G}, \mathcal{H}$ be sub-σ-algebras.

Proposition 5.3. *The conditional expectation satisfies:*

1. ***Linearity:*** $\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$.
2. ***Positivity:*** *If $X \geq 0$ a.s., then $\mathbb{E}[X|\mathcal{G}] \geq 0$ a.s.*

3. ***Taking out what is known**: If Y is $\mathcal{G}$-measurable and $XY \in L^1$, then:*

$$\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}].$$

4. ***Tower Property** (Iterated Conditioning): If $\mathcal{H} \subset \mathcal{G}$ (less information in $\mathcal{H}$), then:*

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}].$$

   Evaluating the best guess of a best guess yields the coarser guess.

5. ***Independence**: If X is independent of $\mathcal{G}$, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.*

6. ***Full Information**: If X is $\mathcal{G}$-measurable, then $\mathbb{E}[X|\mathcal{G}] = X$.*

7. ***Jensen's Inequality**: If $\phi : \mathbb{R} \to \mathbb{R}$ is convex, then $\phi(\mathbb{E}[X|\mathcal{G}]) \leq \mathbb{E}[\phi(X)|\mathcal{G}]$.*

5.3 Geometric Interpretation in L^2

If we restrict ourselves to $L^2(\Omega, \mathcal{F}, \mathbb{P})$, the space is a Hilbert space. The set of $\mathcal{G}$-measurable random variables in L^2, denoted $L^2(\mathcal{G})$, is a closed linear subspace of $L^2(\mathcal{F})$.

Theorem 5.4 (Orthogonal Projection)**.** *For $X \in L^2(\mathcal{F})$, the conditional expectation $\mathbb{E}[X|\mathcal{G}]$ is the **orthogonal projection** of X onto the subspace $L^2(\mathcal{G})$.*

$$\mathbb{E}[X|\mathcal{G}] = argmin_{Z \in L^2(\mathcal{G})} \|X - Z\|_2 = argmin_{Z \in L^2(\mathcal{G})} \mathbb{E}[(X - Z)^2].$$

Using Hilbert space geometry, this means the error $X - \mathbb{E}[X|\mathcal{G}]$ is orthogonal to every vector in the subspace $L^2(\mathcal{G})$:

$$\langle X - \mathbb{E}[X|\mathcal{G}], Z \rangle = 0 \quad \forall Z \in L^2(\mathcal{G}).$$

This interprets the conditional expectation as the "best Mean-Square estimator" of X given the information $\mathcal{G}$.

5.3.1 Example: Best Constant Approximation

Consider the trivial σ-algebra $\mathcal{G} = \{\emptyset, \Omega\}$ (no information).

- The space $L^2(\mathcal{G})$ consists only of **constant random variables** $Z(\omega) = c$.
- The orthogonal projection problem becomes finding the best constant c to approximate X:
$$\min_{c \in \mathbb{R}} \mathbb{E}[(X - c)^2].$$
- By simple calculus (differentiating with respect to c), the minimum is achieved at $c = \mathbb{E}[X]$.
- **Geometric Interpretation**: The expectation $\mathbb{E}[X]$ is the orthogonal projection of the vector X onto the line of constants.
- The Pythagorean theorem holds: $\|X\|^2 = \|\mathbb{E}[X]\|^2 + \|X - \mathbb{E}[X]\|^2$, which corresponds to $\mathbb{E}[X^2] = (\mathbb{E}[X])^2 + \mathrm{Var}(X)$.

5.4 Gaussian Regression

A powerful application of the L^2 projection view is in Gaussian vectors.

Theorem 5.5. *Let $(Y_1, \ldots, Y_n, X)$ be a Gaussian vector in L^2. We want to estimate X given observations $Y = (Y_1, \ldots, Y_n)^T$. The conditional expectation is linear in Y:*

$$\mathbb{E}[X|Y_1, \ldots, Y_n] = \mathbb{E}[X] + \lambda^T(Y - \mathbb{E}[Y])$$

where $\lambda = \Sigma_Y^{-1} Cov(X, Y)$. Here, Σ_Y is the covariance matrix of Y, and $Cov(X, Y)$ is the vector of covariances between X and each Y_i.

5.4.1 Example: Bivariate Normal Case

Let (X, Y) be a bivariate Gaussian vector with means μ_X, μ_Y, variances σ_X^2, σ_Y^2, and correlation ρ.

- We want to estimate X given Y.
- The covariance is $\mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y$.
- The variance of Y is $\Sigma_Y = \sigma_Y^2$.
- Following the theorem, $\lambda = \frac{\rho\sigma_X\sigma_Y}{\sigma_Y^2} = \rho\frac{\sigma_X}{\sigma_Y}$.

Thus, the best estimator is linear:

$$\mathbb{E}[X|Y] = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(Y - \mu_Y).$$

- If $\rho = 0$ (independent), $\mathbb{E}[X|Y] = \mu_X$ (observation Y usesless).
- If $\rho = 1$ (perfect correlation), $\mathbb{E}[X|Y]$ perfectly reconstructs X (up to scaling).

Moreover, the residual error $X - \mathbb{E}[X|Y]$ is independent of $\sigma(Y)$ (a specific property of Gaussianity; essentially orthogonality implies independence for Gaussians).

Chapter 6

Equivalent Probability Measures and Change of Measure

In finance (Girsanov Theorem), we frequently move between the “historical” or “physical” measure $\mathbb{P}$ and a “risk-neutral” measure $\mathbb{Q}$.

6.1 Radon-Nikodym Derivative

Definition 6.1 (Radon-Nikodym Density)**.** Let $\mathbb{P}$ be a reference probability measure. Let Z be a non-negative random variable ($Z \geq 0$ a.s.) with $\mathbb{E}_{\mathbb{P}}[Z] = 1$. We can define a new probability measure $\mathbb{Q}$ by weighing the outcomes by Z:

$$\mathbb{Q}(A) := \int_A Z(\omega)\, d\mathbb{P}(\omega) = \mathbb{E}_{\mathbb{P}}[\mathbb{1}_A Z], \quad \forall A \in \mathcal{F}.$$

The variable Z is called the **Radon-Nikodym derivative** (or density) of $\mathbb{Q}$ with respect to $\mathbb{P}$. We write:

$$Z = \frac{d\mathbb{Q}}{d\mathbb{P}}.$$

Remark 6.1 (Notation: Indicator Function)**.** The symbol $\mathbb{1}_A$ is the **Indicator Function**.

$$\mathbb{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

So $\mathbb{E}_{\mathbb{P}}[\mathbb{1}_A Z]$ is just a fancy way of writing "The integral of Z only over the set A".

Intuition: Multiple Probability Measures? How can we have two probabilities $\mathbb{P}$ and $\mathbb{Q}$?

- Think of a **Coin Toss**.
- $\Omega = \{H, T\}$. The outcomes are fixed.
- Measure $\mathbb{P}$ (Fair Coin): $\mathbb{P}(H) = 0.5, \mathbb{P}(T) = 0.5$.
- Measure $\mathbb{Q}$ (Biased Coin): $\mathbb{Q}(H) = 0.6, \mathbb{Q}(T) = 0.4$.
- We just "re-weighted" the probabilities. The universe Ω didn’t change, only our assignment of likelihoods.
- The factor Z tells us how to re-weight: $Z(H) = \frac{0.6}{0.5} = 1.2$, $Z(T) = \frac{0.4}{0.5} = 0.8$.

6.1.1 Relationship between Expectations

For any random variable X: - $X \in L^1(\mathbb{Q})$ iff $XZ \in L^1(\mathbb{P})$. - The expectation transforms as:

$$\mathbb{E}_{\mathbb{Q}}[X] = \mathbb{E}_{\mathbb{P}}[XZ].$$

Intuition: Why multiply by Z? When we compute an average under $\mathbb{Q}$, we should weight outcomes by their $\mathbb{Q}$-probability. However, if we want to use the $\mathbb{P}$-probability (because we are computing $\mathbb{E}_{\mathbb{P}}$), we must adjust the weight of each outcome ω.

$$\text{Weight in } \mathbb{Q} = Z(\omega) \times \text{Weight in } \mathbb{P}.$$

So, to compute the $\mathbb{Q}$-average of X, we take the $\mathbb{P}$-average of the **re-weighted variable** $X \cdot Z$.

- If $Z(\omega) > 1$, outcome ω is **more likely** under $\mathbb{Q}$ than $\mathbb{P}$, so we give $X(\omega)$ **more weight**.
- If $Z(\omega) < 1$, outcome ω is **less likely** under $\mathbb{Q}$, so we give $X(\omega)$ **less weight**.

6.1.2 Example: Gaussian Shift (Essential for Finance)

This is the building block of the Girsanov Theorem.

- Let $X \sim \mathcal{N}(0,1)$ under $\mathbb{P}$. So $f_{\mathbb{P}}(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$.
- We want to move to a measure $\mathbb{Q}$ where $X \sim \mathcal{N}(\mu, 1)$.
- The target density is $f_{\mathbb{Q}}(x) = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2}$.
- The Radon-Nikodym derivative is the ratio of densities:

$$Z(\omega) = \frac{f_{\mathbb{Q}}(X(\omega))}{f_{\mathbb{P}}(X(\omega))} = \frac{e^{-(X-\mu)^2/2}}{e^{-X^2/2}} = e^{\mu X - \frac{1}{2}\mu^2}.$$

- If we use this Z to re-weight probabilities, X behaves like it has mean μ instead of 0.

6.2 Absolute Continuity and Equivalence

Definition 6.2 (Absolute Continuity)**.** A measure $\mathbb{Q}$ is **absolutely continuous** with respect to $\mathbb{P}$ (denoted $\mathbb{Q} \ll \mathbb{P}$) if $\mathbb{P}(A) = 0 \implies \mathbb{Q}(A) = 0$. Intuitively, "impossible" events under $\mathbb{P}$ must remain "impossible" under $\mathbb{Q}$.

Definition 6.3 (Equivalent Measures)**.** Measures $\mathbb{P}$ and $\mathbb{Q}$ are **equivalent** ($\mathbb{P} \sim \mathbb{Q}$) if $\mathbb{Q} \ll \mathbb{P}$ and $\mathbb{P} \ll \mathbb{Q}$. They agree on which events are possible (have probability > 0) and which are impossible.

Theorem 6.2 (Radon-Nikodym Theorem)**.** *If $\mathbb{Q} \ll \mathbb{P}$, there exists a unique measurable function $Z \geq 0$ such that $d\mathbb{Q} = Zd\mathbb{P}$. If $\mathbb{Q} \sim \mathbb{P}$, then strictly $Z > 0$ almost surely.*

6.3 Conditional Bayes Formula/Kallianpur-Striebel

How does conditional expectation change when we change the measure? This is crucial for pricing (computing $\mathbb{E}^{\mathbb{Q}}[\ldots|\mathcal{F}_t]$).

Theorem 6.3 (Conditional Bayes Formula)**.** *Let $\mathbb{Q} = Z \cdot \mathbb{P}$. Let $\mathcal{G} \subset \mathcal{F}$ be a sub-σ-algebra. For any $X \in L^1(\mathbb{Q})$:*

$$\mathbb{E}_{\mathbb{Q}}[X|\mathcal{G}] = \frac{\mathbb{E}_{\mathbb{P}}[XZ|\mathcal{G}]}{\mathbb{E}_{\mathbb{P}}[Z|\mathcal{G}]}.$$

Intuition: The "local" density on the subspace given by $\mathcal{G}$ is represented by the conditional expectation of Z. We must re-normalize by this local density.